

| Question |     |      |      | Marking details                                                                                                                                                                                                                                                                                                                                      | Marks available |             |          |           |          |          |
|----------|-----|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------|----------|-----------|----------|----------|
|          |     |      |      |                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2         | AO3      | Total     | Maths    | Prac     |
| 6        | (a) |      |      | ${}_{90}^{228}\text{Th} \rightarrow {}_{88}^{224}\text{Ra} + {}_2^4\alpha$ (1)<br>${}_{38}^{90}\text{Co} \rightarrow {}_{39}^{90}\text{Y} + {}_{-1}^0\beta$ (1)                                                                                                                                                                                      |                 | 2           |          | 2         |          |          |
|          | (b) |      |      | Nucleon mass = 90.727 u <b>or</b> nucleon mass +38 e (90.747u) (1)<br>Mass defect attempted with or without electrons 0.84 u or 0.82 u (1)<br>× 931 and division by 90 (1)<br>Answer = 8.69 [MeV per nucleon] (1)<br>If electrons not taken into account answer = 8.47 [MeV per nucleon] award 3 marks<br>782 or 762 [MeV per nucleon] award 2 marks | 1               | 1<br>1<br>1 |          | 4         | 3        |          |
|          | (c) | (i)  |      | Probability of landing on black face = $\frac{1}{4}$ <b>or</b> 0.25 <b>or</b> 25 %                                                                                                                                                                                                                                                                   |                 | 1           |          | 1         | 1        | 1        |
|          |     | (ii) | I.   | Probability of not decaying (i.e. of remaining) after 1 throw = $1 - 0.25 = 0.75$ (1)<br>Probability of remaining after 2 throws = $0.75^2$ <b>or</b> probability of remaining after $n$ throws = $(0.75)^n$ (1)                                                                                                                                     |                 | 2           |          | 2         | 2        | 2        |
|          |     |      | II.  | Number predicted = $N_0 \times (0.75)^n = 31.76 = 32$<br>Accept 31 or 31.76                                                                                                                                                                                                                                                                          |                 | 1           |          | 1         | 1        | 1        |
|          |     |      | III. | Close to 0.75 for many throws <b>or</b> mean close to 0.75 or 32 is close to 35 or fits quite well with $(0.75)^n$ (1)<br>Some further out e.g. 0.90 (1)<br>Random process [these results are to be expected] (1)                                                                                                                                    |                 |             | 3        | 3         |          | 3        |
|          |     |      |      | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                              | <b>1</b>        | <b>9</b>    | <b>3</b> | <b>13</b> | <b>7</b> | <b>7</b> |